

## MATH 454: Survey of Analysis

Homework 1

Due September 4

1. Is the collection of all finite subsets of  $\mathbb{R}$  an algebra of sets? Why or why not?
2. Prove that if  $\mathcal{C}$  is a collection of subsets of  $X$ , there is a smallest  $\sigma$ -algebra  $\mathcal{A}$  that contains  $\mathcal{C}$ . This is Proposition 1.3 in the notes.
3. Prove that the outer measure  $\mathcal{L}^*$  is translation invariant, meaning that  $\mathcal{L}^*(A) = \mathcal{L}^*(A + y)$  for any subset  $A \subseteq \mathbb{R}$  and  $y \in \mathbb{R}$ .
4. Show that if  $\mathcal{L}^*(A) = 0$ , then for any subset  $B$  of  $\mathbb{R}$  it is true that  $\mathcal{L}^*(A \cup B) = \mathcal{L}^*(B)$ .